Summary Statistics of the Fannie Mae Single-Family Loan Performance Data

2026-11-01

These are the quarters that I have cleaned so far. I only use the primary dataset, which is not every loan Fannie Mae acquired (see the docs). The recent vintages have had little time to default, so compare outcomes at the same loan age.

Rows read and loans per quarter

```
log <- read_csv(log_file)

log %>%
  select(Quarter = quarter, `Loan-months` = rows_read, Loans = n_loans,
         `Last period` = last_act_period, `Missing FICO` = share_missing_fico) %>%
  knitr::kable(format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Quarter	Loan-months	Loans	Last period	Missing FICO
2007Q1	16,804,063	253,279	2026-06	0.0013
2010Q1	21,032,502	323,174	2026-06	0.0008
2018Q1	23,902,313	458,830	2026-06	0.0005
2021Q1	80,080,858	1,413,390	2026-06	0.0002
2023Q1	7,686,914	210,810	2026-06	0.0008

Checks

Check for loans with duplicate months, more than one zero balance code, or more than one MSA, and how often `orig_classic_fico` equals the lower of the two older scores.

```
log %>%
  select(Quarter = quarter, `Dup. months` = n_loans_dup_months,
         `>1 zero bal. code` = n_loans_multi_zb, `>1 MSA` = n_loans_multi_msa,
         `Both FICO` = fico_both_n, `Share equal` = fico_equal_share) %>%
  knitr::kable(format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Quarter	Dup. months	>1 zero bal. code	>1 MSA	Both FICO	Share equal
2007Q1	0	0	5,952	7,546	1
2010Q1	0	0	13,507	16,991	1
2018Q1	0	0	40,637	116,382	1
2021Q1	0	0	399,828	1,115,256	1
2023Q1	0	0	49,471	167,223	1

Loan characteristics by acquisition quarter

```
cols <- c("acq_quarter", "orig_upb", "oltv", "dti", "orig_rate", "fico_orig",
          "terminated", "prepaid", "credit_event", "default_completed",
          "ever_dlq90", "ever_modified")

loans <- map_dfr(loan_files, ~ read_dta(.x, col_select = all_of(cols)))

loans %>%
  group_by(Quarter = acq_quarter) %>%
  summarise(Loans = n(),
            `Mean UPB` = mean(orig_upb, na.rm = TRUE),
            `Mean OLTV` = mean(oltv, na.rm = TRUE),
            `Mean DTI` = mean(dti, na.rm = TRUE),
            `Mean rate` = mean(orig_rate, na.rm = TRUE),
            `Mean FICO` = mean(fico_orig, na.rm = TRUE)) %>%
  knitr::kable(digits = 1, format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Quarter	Loans	Mean UPB	Mean OLTV	Mean DTI	Mean rate	Mean FICO
2007Q1	253,279	192,779.9	69.2	37.6	6.2	717.4
2010Q1	323,174	218,182.4	65.3	33.4	4.9	758.4
2018Q1	458,830	234,667.2	74.8	36.4	4.3	741.6
2021Q1	1,413,390	282,747.7	67.0	32.9	2.8	760.7
2023Q1	210,810	312,828.6	75.3	37.5	6.3	749.0

Outcomes by acquisition quarter (share of loans)

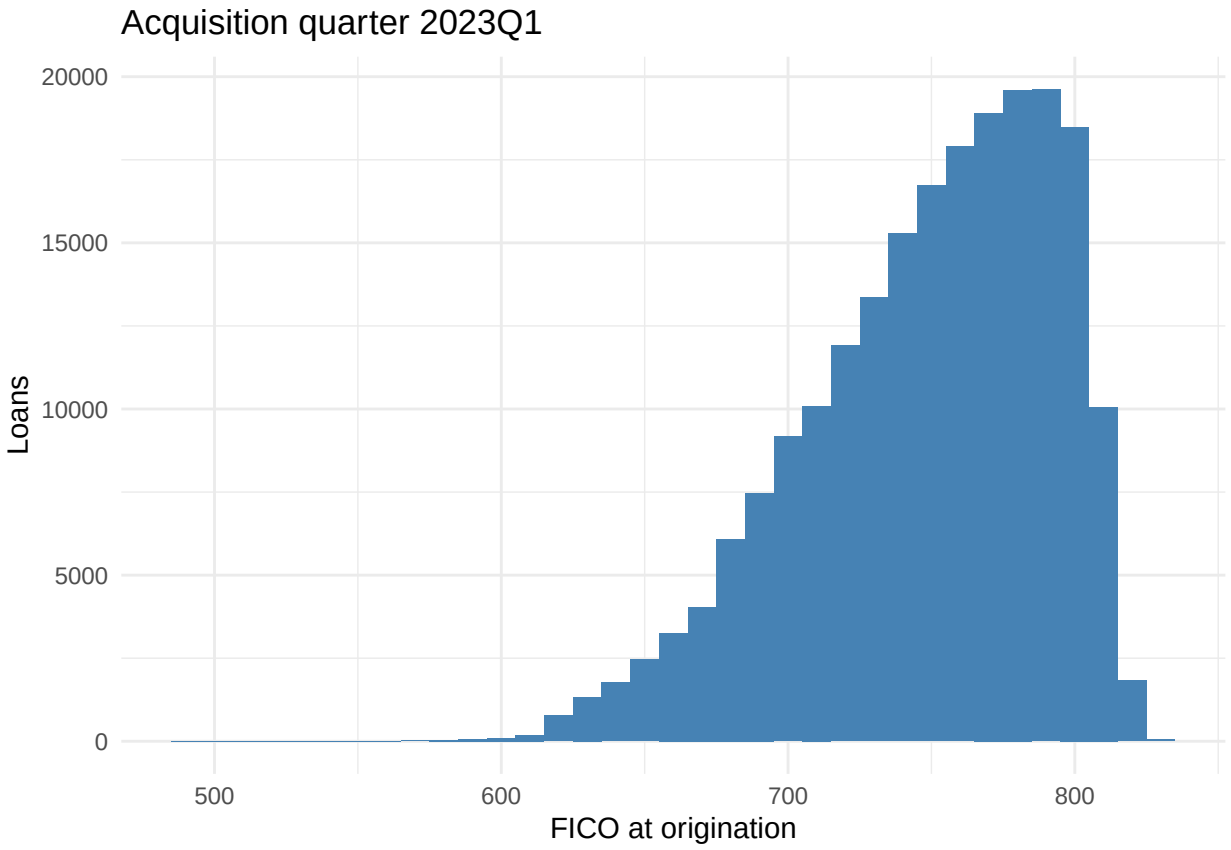
```
loans %>%
  group_by(Quarter = acq_quarter) %>%
  # ever_dlq90 is missing for the few loans whose only status is unknown (XX)
  summarise(Terminated = mean(terminated, na.rm = TRUE), Prepaid = mean(prepaid, na.rm = TRUE),
            `Credit event` = mean(credit_event, na.rm = TRUE),
            `Default (completed)` = mean(default_completed, na.rm = TRUE),
            `Ever 90+ days` = mean(ever_dlq90, na.rm = TRUE),
            `Ever modified` = mean(ever_modified, na.rm = TRUE)) %>%
  knitr::kable(digits = 3) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Quarter	Terminated	Prepaid	Credit event	Default (completed)	Ever 90+ days	Ever modified
2007Q1	0.972	0.863	0.086	0.085	0.164	0.071
2010Q1	0.950	0.941	0.007	0.006	0.027	0.008
2018Q1	0.757	0.752	0.002	0.002	0.066	0.014
2021Q1	0.232	0.230	0.000	0.000	0.016	0.004
2023Q1	0.266	0.261	0.002	0.001	0.029	0.009

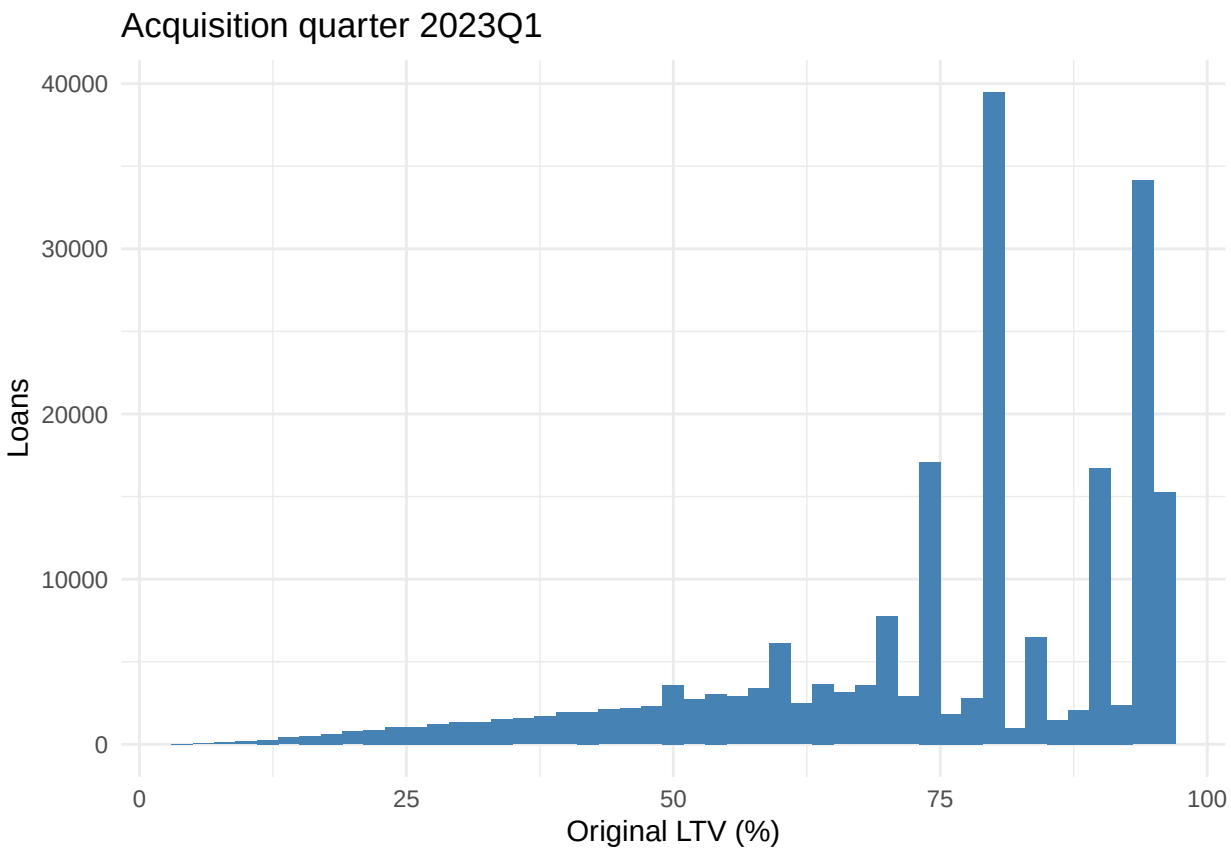
Credit score and LTV at origination (the latest quarter I've cleaned)

```
last_q <- max(loans$acq_quarter)
```

```r
loans %>%
  filter(acq_quarter == last_q) %>%
  ggplot(aes(x = fico_orig)) +
    geom_histogram(binwidth = 10, fill = "steelblue", na.rm = TRUE) +
    labs(x = "FICO at origination", y = "Loans", title = paste("Acquisition quarter", last_q)) +
    theme_minimal()
```




```r
loans %>%
  filter(acq_quarter == last_q) %>%
  ggplot(aes(x = oltv)) +
    geom_histogram(binwidth = 2, fill = "steelblue", na.rm = TRUE) +
    labs(x = "Original LTV (%)", y = "Loans", title = paste("Acquisition quarter", last_q)) +
    theme_minimal()
```

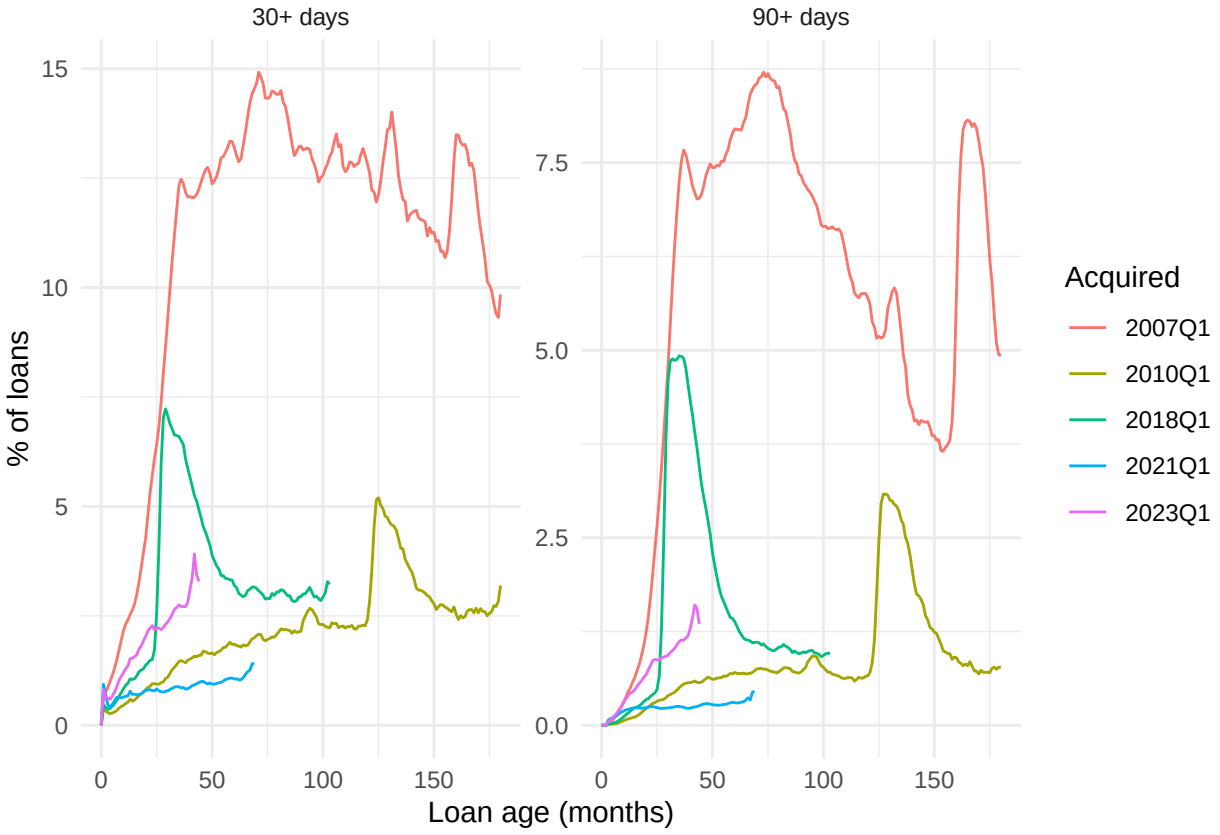


Delinquency by loan age

Share of loans that are 30+ and 90+ days delinquent at each age, by acquisition quarter. Later ages only include loans that are still reporting.

```
dlq <- read_csv(here("outputs", "fannie_mae_dlq_by_age.csv"))

dlq %>%
  filter(n_known >= 1000) %>%
  pivot_longer(c(share_dlq30, share_dlq90), names_to = "type", values_to = "share") %>%
  mutate(type = recode(type, share_dlq30 = "30+ days", share_dlq90 = "90+ days")) %>%
  ggplot(aes(x = loan_age, y = 100 * share, color = acq_quarter)) +
    geom_line() +
    facet_wrap(~ type, scales = "free_y") +
    labs(x = "Loan age (months)", y = "% of loans", color = "Acquired") +
    theme_minimal()
```



MSA x year (This would be how I would link to HMDA)

```
msa_year <- read_dta(here("data", "clean", "fannie-mae", "fannie_mae_msa_year.dta"))

msa_year %>%
  group_by(Year = orig_year) %>%
  summarise(MSAs = n(), Loans = sum(n_loans), `Mean FICO` = mean(fico_mean, na.rm = TRUE),
            `Mean OLTV` = mean(oltv_mean, na.rm = TRUE)) %>%
  knitr::kable(digits = 1, format.args = list(big.mark = ",")) %>%
  kableExtra::kable_styling(latex_options = c("striped", "hold_position"))
```

Year	MSAs	Loans	Mean FICO	Mean OLTV
1,999	3	10	625.4	75.3
2,001	16	84	717.8	66.7
2,002	17	101	724.3	74.5
2,003	16	52	719.4	73.2
2,004	13	39	747.3	77.2
2,005	98	464	729.9	67.3
2,006	406	126,574	718.5	70.6
2,007	406	125,956	716.0	70.1
2,008	17	51	754.2	68.4
2,009	406	155,258	753.9	67.5
2,010	406	167,864	755.8	66.6
2,013	1	1	795.0	61.0

2,016	19	37	751.2	68.4
2,017	394	198,142	741.5	77.1
2,018	394	260,652	741.8	76.6
2,019	84	230	761.7	71.9
2,020	393	417,656	756.8	71.8
2,021	393	995,736	757.3	70.3
2,022	392	74,990	745.4	75.6
2,023	393	135,586	749.2	75.3